

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks: 20

Annexure A

Descriptive Written Test

Code of Conduct:

I M. SHAKTHI 192312538 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

20+20+22+18+12
= 92
10

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

Descriptive TestSmart home automation system using intelligent agents:

A smart home automation system controls lighting, temperature, security based on user behaviour, environmental conditions. It receives real time inputs from sensors such as motion detectors, temperature sensors, light sensors, clock.

Types of Intelligent agents and their functions:

1) Simple reflex agent: A simple reflex agent acts only on the current input using predefined IF-THEN rules without considering past information

Working: If motion is detected + Turn on lights
If room is empty - Turn off lights

Advantages:

- * fast response
- * easy to design
- * suitable for simple tasks

Disadvantages:

- * cannot remember previous events
- * Cannot learn user preferences
- * perform poorly in complex situation

2) Model Based reflex agent: A model based agent maintains an internal model of the environment & uses both current and previous information.

Working: Remembers whether someone is already inside house, knows if lights were previously on/off, detects doors were open.

Advantages:

- * Handles partially observable environments
- * Makes better decisions
- * Improves system reliability

Disadvantages:

- * requires memory
- * more computationally expensive

3) Goal based agent: A goal based agent chooses actions to help achieve a specific goal.

goals in smart home:

- * maintain comfortable room temperature
- * minimize electricity consumption.
- * ensure home security
- * Keep indoor lighting

Working:

ex: goal: save energy

action: turn off unused appliances, reduce AC usage, close curtains during daytime.

advantages:

- * Intelligent planning
- * flexible decision-making
- * can solve complex problems

disadvantages:

- * slower than reflex agents.
- * requires planning algorithms.

4) utility based agent: A utility based agent selects the action that provides the highest overall benefit by evaluating multiple factors.

utility factors:

- * user comfort
- * energy efficiency
- * security
- * cost

Advantages:

- * produces optimal decisions
- * balances multiple objectives
- * suitable for real world smart homes

Working: Example: If electricity cost is high, increase AC temperature slightly, use natural lighting.

Disadvantages: * complex design.

- * requires utility calculations.

1) approach: Hybrid intelligent agent

- * A Hybrid agent combining model based, goal-based, utility based agents is the most effective solution.
- * It uses memory to understand environmental conditions.
- * Plans actions to achieve comfort & security.
- * selects the best action based on utility.
- * learns user preferences, responds quickly to emergencies.

2) Robot navigation in an unknown maze using uninformed search:

* A robot is placed in an unknown maze and must find the exit without prior knowledge of the maze. Since no heuristic information is available, uninformed search algorithms such as uniform cost search (UCS) & iterative deepening search (IDS) are used.

1) Uniform Cost search (UCS)

It expands the node with the lowest cumulative path cost first. It guarantees the least cost path.

working:

- * Start from the initial position.
- * Store nodes in a priority queue.
- * expand the node with the smallest cost.
- * continue until the exit is found.

advantages:

- * Complete
- * optimal
- * suitable for weighted paths

Disadvantages:

- * High memory usage
- * slow for large search spaces

Complexity:

Time complexity: $O(b^{1 + \lceil C^*/\epsilon \rceil})$

Space complexity: $O(b^{1 + \lceil C^*/\epsilon \rceil})$

where, b = branching factor

C^* = optimal path cost

ϵ = mini step cost

2) Iterative deepening search (IDS)

It repeatedly performs depth first search with increasing depth limits until the goal is found.

Working: * Search depth 0

* Search depth 1

* Search depth 2

* continue increasing the depth until the exit is reached.

Advantages:

* requires very little memory

* complete

* suitable when soln depth is unknown

Disadvantages:

* repeats node expansions

* not optimal when path costs differ.

Complexity: Time: $O(b^d)$

Space: $O(b^d)$

where, b = branching factor

d = depth of solution

For robot navigation in an unknown maze, both UCS and IDS are useful uninformed search algorithms. UCS provides the optimal path when path costs vary, while IDS is memory efficient for unknown depth problems.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$20 + 20 + 25 + 20 + 15 = 100$$